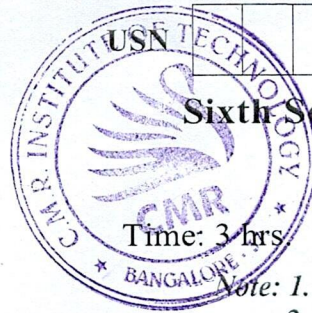




Sixth Semester B.E./B.Tech. Degree Examination, June/July 2025 Machine Learning

BCS602

Time: 3 hrs

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C																														
Q.1	a.	State Tom Mitchell’s definition of machine learning. List and explain the challenges of machine learning.	7	L1	CO1																														
	b.	List and explain the visualization aids available for univariate data analysis with example for each.	7	L2	CO1																														
	c.	For the patients age list {12, 14, 19, 22, 24, 26, 28, 31, 34}. Find the IQR.	6	L3	CO1																														
OR																																			
Q.2	a.	Explain in detail the machine learning process with a neat diagram.	7	L2	CO1																														
	b.	Explain data preprocessing with measures to solve the problem of missing data.	7	L2	CO1																														
	c.	Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9} and plot the box plot for the same.	6	L3	CO1																														
Module – 2																																			
Q.3	a.	Let the data points be $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$. Apply Principal Component Analysis (PCA) and find the transformed data.	10	L3	CO1																														
	b.	Apply candidate elimination algorithm on the dataset given in Table Q.3(b) to obtain the complete version space. Table Q.3(b) <table><tr><td>CGPA</td><td>Interactiveness</td><td>Practical knowledge</td><td>Communication skills</td><td>Logical thinking</td><td>Job offer</td></tr><tr><td>≥ 9</td><td>Yes</td><td>Excellent</td><td>Good</td><td>Fast</td><td>YES</td></tr><tr><td>≥ 9</td><td>Yes</td><td>Good</td><td>Good</td><td>Fast</td><td>YES</td></tr><tr><td>≥ 8</td><td>No</td><td>Good</td><td>Good</td><td>Fast</td><td>NO</td></tr><tr><td>≥ 9</td><td>Yes</td><td>Good</td><td>Good</td><td>Slow</td><td>YES</td></tr></table>	CGPA	Interactiveness	Practical knowledge	Communication skills	Logical thinking	Job offer	≥ 9	Yes	Excellent	Good	Fast	YES	≥ 9	Yes	Good	Good	Fast	YES	≥ 8	No	Good	Good	Fast	NO	≥ 9	Yes	Good	Good	Slow	YES	10	L3	CO2
CGPA	Interactiveness	Practical knowledge	Communication skills	Logical thinking	Job offer																														
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≥ 8	No	Good	Good	Fast	NO																														
≥ 9	Yes	Good	Good	Slow	YES																														
OR																																			
Q.4	a.	Find Singular Value Decomposition (SVD) of the matrix $A = \begin{pmatrix} 1 & 2 \\ 4 & 9 \end{pmatrix}$.	10	L3	CO2																														

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	b.	Write Find-S algorithm. Apply the algorithm to obtain the hypothesis for the dataset given in the Table Q.4(b).	10	L3	CO2																																				
Table Q.4(b) <table> <tr> <td>Sky</td> <td>Air temp</td> <td>Humidity</td> <td>Wind</td> <td>Water</td> <td>Forecast</td> <td>Enjoy sport</td> </tr> <tr> <td>Sunny</td> <td>Warm</td> <td>Normal</td> <td>Strong</td> <td>Warm</td> <td>Same</td> <td>YES</td> </tr> <tr> <td>Sunny</td> <td>Warm</td> <td>High</td> <td>Strong</td> <td>Warm</td> <td>Same</td> <td>YES</td> </tr> <tr> <td>Rainy</td> <td>Cold</td> <td>High</td> <td>Strong</td> <td>Warm</td> <td>Change</td> <td>NO</td> </tr> <tr> <td>Sunny</td> <td>Warm</td> <td>High</td> <td>Strong</td> <td>Cool</td> <td>Change</td> <td>YES</td> </tr> </table>			Sky	Air temp	Humidity	Wind	Water	Forecast	Enjoy sport	Sunny	Warm	Normal	Strong	Warm	Same	YES	Sunny	Warm	High	Strong	Warm	Same	YES	Rainy	Cold	High	Strong	Warm	Change	NO	Sunny	Warm	High	Strong	Cool	Change	YES				
Sky	Air temp	Humidity	Wind	Water	Forecast	Enjoy sport																																			
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Sunny	Warm	High	Strong	Cool	Change	YES																																			
Module – 3																																									
Q.5	a.	Apply K-nearest neighbor algorithm, for the dataset given in Table Q.5(a). Given a test instance (6.1, 40, 5), use the training set to classify the test instance. Choose K = 3.	6	L3	CO3																																				
Table Q.5(a) <table> <tr> <td>CGPA</td> <td>Assessment</td> <td>Project submitted</td> <td>Result</td> </tr> <tr> <td>9.2</td> <td>85</td> <td>8</td> <td>PASS</td> </tr> <tr> <td>8</td> <td>80</td> <td>7</td> <td>PASS</td> </tr> <tr> <td>8.5</td> <td>81</td> <td>8</td> <td>PASS</td> </tr> <tr> <td>6</td> <td>45</td> <td>5</td> <td>FAIL</td> </tr> <tr> <td>6.5</td> <td>50</td> <td>4</td> <td>FAIL</td> </tr> <tr> <td>5.8</td> <td>38</td> <td>5</td> <td>FAIL</td> </tr> </table>			CGPA	Assessment	Project submitted	Result	9.2	85	8	PASS	8	80	7	PASS	8.5	81	8	PASS	6	45	5	FAIL	6.5	50	4	FAIL	5.8	38	5	FAIL											
CGPA	Assessment	Project submitted	Result																																						
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8	80	7	PASS																																						
8.5	81	8	PASS																																						
6	45	5	FAIL																																						
6.5	50	4	FAIL																																						
5.8	38	5	FAIL																																						
	b.	Explain types of regression methods and limitations of regression methods.	7	L2	CO3																																				
	c.	Explain the structure of a decision tree and write the procedure to construct a decision the using ID3 algorithm.	7	L2	CO3																																				
OR																																									
Q.6	a.	Write the nearest-centroid classifier algorithm. Apply the same to predict the class for the given test instance (6, 5) using the training dataset given in Table Q.6(a).	7	L3	CO3																																				
Table Q.6(a) <table> <tr> <td>X</td> <td>Y</td> <td>Class</td> </tr> <tr> <td>3</td> <td>1</td> <td>A</td> </tr> <tr> <td>5</td> <td>2</td> <td>A</td> </tr> <tr> <td>4</td> <td>3</td> <td>A</td> </tr> <tr> <td>7</td> <td>6</td> <td>B</td> </tr> <tr> <td>6</td> <td>7</td> <td>B</td> </tr> <tr> <td>8</td> <td>5</td> <td>B</td> </tr> </table>			X	Y	Class	3	1	A	5	2	A	4	3	A	7	6	B	6	7	B	8	5	B																		
X	Y	Class																																							
3	1	A																																							
5	2	A																																							
4	3	A																																							
7	6	B																																							
6	7	B																																							
8	5	B																																							
	b.	Distinguish between i) Regression and correlation ii) Regression and causation iii) Linearity and non-linearity relationships.	6	L2	CO3																																				
	c.	Explain the advantages and disadvantages of decision tree. Write the general algorithm for decision tree.	7	L2	CO3																																				

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Module – 4

Q.7	a.	Using Naïve bayes classifier classify the new data (Red, SUV, Domestic) using the training dataset given in Table Q.7(a). Table Q.7(a) <table><tr><td>Color</td><td>Type</td><td>Origin</td><td>Stolen</td></tr><tr><td>Red</td><td>Sports</td><td>Domestic</td><td>YES</td></tr><tr><td>Red</td><td>Sports</td><td>Domestic</td><td>NO</td></tr><tr><td>Red</td><td>Sports</td><td>Domestic</td><td>YES</td></tr><tr><td>Yellow</td><td>Sports</td><td>Domestic</td><td>NO</td></tr><tr><td>Yellow</td><td>Sports</td><td>Imported</td><td>YES</td></tr><tr><td>Yellow</td><td>SUV</td><td>Imported</td><td>NO</td></tr><tr><td>Yellow</td><td>SUV</td><td>Imported</td><td>YES</td></tr><tr><td>Yellow</td><td>SUV</td><td>Domestic</td><td>NO</td></tr><tr><td>Red</td><td>SUV</td><td>Imported</td><td>NO</td></tr><tr><td>Red</td><td>Sports</td><td>Imported</td><td>YES</td></tr></table>	Color	Type	Origin	Stolen	Red	Sports	Domestic	YES	Red	Sports	Domestic	NO	Red	Sports	Domestic	YES	Yellow	Sports	Domestic	NO	Yellow	Sports	Imported	YES	Yellow	SUV	Imported	NO	Yellow	SUV	Imported	YES	Yellow	SUV	Domestic	NO	Red	SUV	Imported	NO	Red	Sports	Imported	YES	10	L3	CO4
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	b.	Explain the simple model of an artificial neuron along with the artificial neural network structure.	10	L2	CO4																																												

OR

Q.8	a.	Explain Bayes theorem, Maximum A Posteriori (MAP) hypothesis and Maximum Likelihood (ML) hypothesis in detail.	10	L2	CO4
	b.	Explain different activation functions used in artificial neural network.	10	L2	CO4

Module – 5

Q.9	a.	Consider the following set of data given in Table Q.9(a). Cluster it using K-means algorithm with initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds. Table Q.9(a)<table><tr><td>Objects</td><td>X-coordinate</td><td>Y-coordinate</td></tr><tr><td>1</td><td>2</td><td>4</td></tr><tr><td>2</td><td>4</td><td>6</td></tr><tr><td>3</td><td>6</td><td>8</td></tr><tr><td>4</td><td>10</td><td>4</td></tr><tr><td>5</td><td>12</td><td>4</td></tr></table>	Objects	X-coordinate	Y-coordinate	1	2	4	2	4	6	3	6	8	4	10	4	5	12	4	10	L3	CO5
Objects	X-coordinate	Y-coordinate																					
1	2	4																					
2	4	6																					
3	6	8																					
4	10	4																					
5	12	4																					
	b.	Explain the various components of reinforcement learning.	10	L2	CO5																		

OR

Q.10	a.	Find the Manhattan and Chebyshev distance if the coordinates of the objects are (0, 3) and (5, 8).	4	L3	CO5
	b.	Explain the mean shift clustering algorithm.	6	L2	CO5
	c.	List and explain the i) Characteristics of reinforcement learning ii) Challenges of reinforcement learning iii) Applications of reinforcement learning	10	L3	CO5
